

CSE 4113 – Internet Programming Lab

Project Report

Hopper

Team: Four-Eyed Raven

Submitted By

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TODO: Add member 4 name	TODO: Add student ID
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TODO: Add submission date

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Project Links

Item	URL
Public Git Repository	TODO: Add public repository URL if available
Deployed Application URL	TODO: Add public live application URL
API Docs (Swagger/Postman) – public link	TODO: Add public API documentation URL
Demo Video	TODO: Add public demo video URL

1. Introduction

1.1 Problem Definition & Context

TODO: Summarize the problem Hopper solves using verified project goals from the repository.

1.2 Target Users & Use Cases

TODO: List verified user groups, roles, and their core use cases.

1.3 Core Features

TODO: Document the confirmed feature set from the current repository and deployment flow.

1.4 Tech Stack Overview

TODO: Summarize the verified frontend, backend, database, messaging, authentication, and infrastructure stack.

2. System Architecture

2.1 High-Level Architecture Diagram

TODO: Insert a system architecture diagram exported to `report/figures/`.

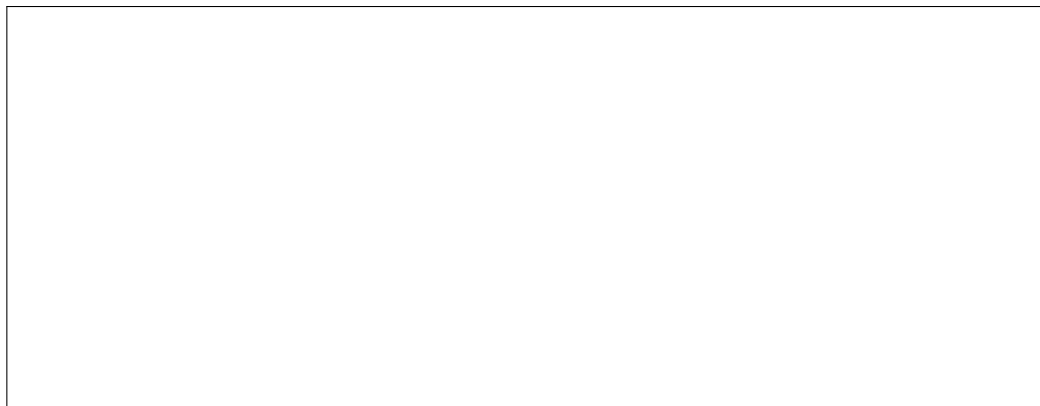


Figure 1: Placeholder for a Hopper architecture diagram from `report/figures/`.

2.2 Frontend Architecture

TODO: Describe the verified frontend structure, routes, and state management choices.

2.3 Backend Architecture

TODO: Describe the verified backend services, communication patterns, and design choices.

2.4 Database Design

TODO: Add the ERD, schema summary, indexing decisions, and verified real-time update flow.

3. API Documentation

3.1 API Design Overview

TODO: State the verified API style, base paths, and any versioning decisions.

3.2 Endpoint Reference

TODO: Replace the placeholder rows below with verified endpoints only.

Method	Route	Description	Auth Required
TODO: GET	TODO: /example	TODO: Add verified endpoint details	TODO: Yes/No
TODO: POST	TODO: /example	TODO: Add verified endpoint details	TODO: Yes/No
TODO: PUT	TODO: /example	TODO: Add verified endpoint details	TODO: Yes/No

3.3 Swagger / Postman Collection Link

TODO: Add the public Swagger UI or Postman collection URL if available.

3.4 Sample Request & Response

TODO: Insert one verified request/response pair for a representative endpoint.

Listing 1: Placeholder for a sample API request

```
# TODO: Add a verified request example.
```

4. Authentication & Security

4.1 Authentication Strategy

TODO: Describe the verified authentication flow and why it was chosen.

4.2 Authorization & Role Management

TODO: Document the confirmed application roles and access-control model.

4.3 Security Measures

TODO: Summarize verified validation, secrets handling, rate limiting, network, and transport protections.

4.4 Known Vulnerabilities & Mitigations

TODO: Add verified known risks, mitigations, and remaining gaps from the repository or test docs.

5. Testing & Quality Assurance

5.1 Testing Strategy

TODO: Summarize the verified unit, integration, end-to-end, and other automated test layers.

5.2 Test Coverage Summary

TODO: Add measured coverage or coverage-report references after verification.

5.3 Bug Tracking & Resolution Log

TODO: Summarize verified bug-tracking artifacts such as BUG_TRACKING_LOG.md.

5.4 Sample Test Cases

TODO: Add representative verified test cases from the current suites.

6. CI/CD & Deployment

6.1 Pipeline Overview

TODO: Describe the verified CI and delivery workflow from repository automation and scripts.

6.2 Environments

TODO: Document the confirmed development, test, staging, and production environments if they exist.

6.3 Deployment Steps

TODO: Summarize the verified deployment procedure from current infrastructure files.

6.4 Hosting Platform & Live URL

TODO: Add the actual hosting platform and public live URL after verification.

6.5 Monitoring & Logging

TODO: Describe the verified monitoring, logging, and observability components.

7. Repository & Documentation Quality

7.1 Branching Strategy & Commit Conventions

TODO: Add the actual branching and commit practices only after verifying repository history or team policy.

7.2 README Completeness Checklist

TODO: Audit the README against setup, environment, run/test, and usage expectations.

7.3 Code Organization / Folder Structure

TODO: Summarize the verified repository structure and service boundaries.

7.4 Local Setup Instructions

TODO: Condense the verified local setup steps for reproducibility.

8. Evaluation & Reflection

8.1 Web Performance & Core Web Vitals

TODO: Add measured Lighthouse or browser-tool performance metrics later.

8.2 Challenges & Solutions

TODO: Add verified implementation or deployment challenges and how they were addressed.

8.3 Limitations & Future Work

TODO: Add realistic limitations and future improvements based on verified project scope.

8.4 Lessons Learned

TODO: Summarize team-level lessons after the full report content is drafted.

8.5 Individual Responsibility

TODO: Replace the placeholder rows with actual team-member responsibilities and contribution evidence.

Member Name	Core Responsibilities & Feature Ownership	Key PRs / Commit Contributions	Estimated % Contribution
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %
TODO: Add name	TODO: Add responsibilities	TODO: Add PRs or commits	TODO: Add %

A. Screenshots / UI Walkthrough

TODO: Add annotated screenshots, UI flows, and captions using image files placed under report/figures/.

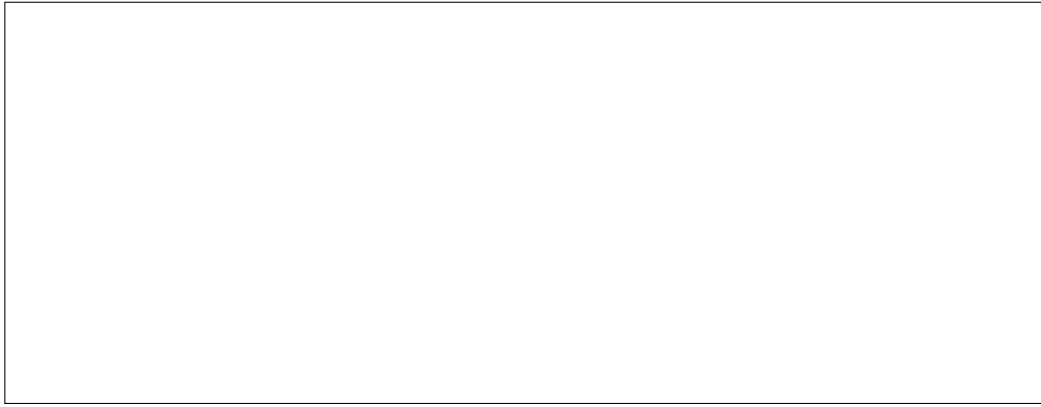


Figure 2: Placeholder for a UI walkthrough screenshot loaded from `report/figures/`.